/*Check a number is Armstrong or not.

A number is called an Armstrong number if the sum of each digit raised to the power of the number of digits is equal to the original number.

153 1634

*/

Three Digit Armstrong number :

$153 = 1^3 + 5^3 + 3^3$

$1 + 125 + 27 = 153$

Four Digit Armstrong number:

1634 [Total number of digits = 4]

$1^4 + 6^4 + 3^4 + 4^4$

Math class :

Math is a predefined utility class (Helper class) available in java.lang pacakge.

It provides various static methods to perform numeric operation.

public static double pow(double x, double y) : It is used to find out the power of the given number i.e x^y

Example :

```
void main()
{
     double x = 3.0;
     double y = 4.0;
     double value = Math.pow(x, y);
     IO.println(value); //81.0
}
```

```
//Program to find out the armstrong number
/*
Logic for Armstrong  numer :
-----------------------------
1) Initialize digits = 0 and double sum = 0.0. [digits = total number of digits]
2) Store the number in a temporary variable i.e original
3) count the total digits till num!=0 (digits++)
4) Calculate digit to the power and add it to a sum. [sum = sum + Math.pow(digit, digits);]
5) Remove the last digit using / 10.
6) After processing all digits, compare the sum with the original number.
If equal, it is an Armstrong Number.
*/


 void main()
 {
     int number = Integer.parseInt(IO.readln("Enter a number :"));
     getArmstrong(number);
 }

 void getArmstrong(int num)
 {
  int original = num;

  //count the number of digits
  int digits = 0; //3 [Total number of digits]
  int temp = num;

   while(temp !=0)   //temp = 0
     {
      temp = temp /10;
      digits++;
     }

    //finding the sum
    double sum = 0;

      while(num!=0)   //num =
      {
           int digit = num % 10;   //digit = 5
           sum = sum + Math.pow(digit, digits);
       num = num /10;
      }

    if(original == sum)
      {
     IO.println(original +" is an armstrong number");
      }
      else
      {
           IO.println(original +" is not an armstrong number");
      }

 }
```

```
/* Perfect Number
A perfect number is a positive integer that equals the sum of its proper positive divisors, excluding the number itself.

Example :
a)  6  -> 1, 2 , 3   so [1 + 2 + 3 = 6]

b)  28 -> 1, 2, 4, 7, 14 [1 + 2 + 4 + 7 + 14 = 28]

*/

 void main()
 {
     int number = Integer.parseInt(IO.readln("Enter a number :"));
     IO.println("Is "+number+" perfect number : "+isPerfect(number));
 }

 boolean isPerfect(int num)  //28
 {
     if(num <=0)
     {
          IO.println("Invalid number");
          return false;
     }

   int sum = 0;
     int temp = num;

     for(int i=1; i<=num/2; i++)
     {
      if(num % i ==0)
          {
             sum = sum + i;
          }
     }
   return temp==sum;
 }
```

```
/*
Logic for Perfect number :
--------------------------
1) Store the number in a temp variable
2) Initialize sum = 0.
3) Find all factors from 1 to num/2.
4) If num % i == 0, add i to sum.
5) After the loop, compare sum with temp.
6) If equal, it is a Perfect Number.
*/
```

```
 //count all the even numbers in a given digit
 void main()
 {
      int number = Integer.parseInt(IO.readln("Enter a number :"));
      IO.println("Total even digits are :"+countEvenDigits(number));
 }

 int countEvenDigits(int num)
 {
      int count = 0;

      while(num !=0)
      {
           int digit = num % 10;

           if(digit % 2==0)
           {
                count++;
           }

           num = num /10;
      }
      return count;
 }
```

Can we write multiple methods with same name ?

Yes, we can write multiple methods with same name but parameter must be different otherwise code will not compile. [Known as Method Overloading]

Note: - We can also write multiple main methods with different parameter but JVM will always execute the main method which takes String [] args (String array) as a parameter as shown in the program below.

Case 1:

```
void main ()
{
     IO.println ("Hello India");
}


public static void main(String [] args)
{
     IO.println("Hello World");
}
```

Output : Hello World

Case 2 :

```
void main(String [] args)
{
     IO.println("Hello India");
}


void main()
{
     IO.println("Hello World");

}
```

Output : Hello India

Case 3 :

```
void main(int [] args)
{
     IO.println("Hello India");
}


void main()
{
     IO.println("Hello World");

}
```

Output : Hello World

So, final conclusion, Java provides the priority to the main() method using following order :

a) public static void main(String [] args)

b) void main(String [] args)

c) void main()

Note: we cannot write a and b together due to ambiguity. Both are accepting String [] args as a parameter

String [] args:

String is a predefined class and args is an array variable of type String so, it can hold multiple values.

IQ:

Why the main method of java accepts String array as a parameter?

String is a collection of alpha-numeric character so it can accept all different kind of values. Java software people has provided String array as a parameter so it can ACCEPT **MULTIPLE VALUES OF DIFFERENT TYPE**, that means providing more wider scope to accept heterogeneous types of values.